



**SRI SRI UNIVERSITY**  
FACULTY OF ENGINEERING & TECHNOLOGY

**PROJECT REPORT**  
**on**  
***cs shield: defensive monitoring system***  
***for scada networks***  
**Bachelor Of Computer Science and Technology**  
**(CSCD)**

**SUBMITTED BY:-**

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## ***Certificate Of Submission***

This is to certify that Nisha Priya, Shivam Raj, Aloukik Agrawal and Ved Patil enrolled in B.Tech Computer Science and Engineering at Sri Sri University have successfully completed and submitted their major project titled "***cs shield: defensive monitoring system for scada networks***" as part of their academic curriculum. The project was submitted on 24th of April, 2026.

Throughout the duration of this project, the students have demonstrated a commendable understanding of the subject matter and have exhibited exceptional skills in research, analysis, and presentation.

We acknowledge the effort and dedication put forth by the students in completing his project and commend their commitment to academic excellence.

Signature of The Mentor

## *Acknowledgement*

We would like to express our sincere gratitude to all those who contributed to the success of this project. First and foremost, we extend our heartfelt thanks to the management and Experts team for their continuous support and encouragement throughout the project.

We are immensely grateful to our team members for their dedication, hard work, and collaboration, which played a pivotal role in developing and implementing the predictive maintenance model. Their expertise, creativity, and commitment have been instrumental in achieving our project objectives.

We would like to acknowledge the invaluable support and guidance provided by our mentor (Ms.Ishita Kosambia),her insights, feedback, and expertise have significantly enriched the project and enhanced its impact.

Last but not least, we acknowledge the support of our families, friends, and colleagues who stood by us and provided encouragement and understanding throughout the project journey.

Thank you to everyone involved for their contributions, dedication, and commitment to excellence, making this project a success.

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# **CHAPTER 1**

## **INTRODUCTION**

### **1.1 Background of SCADA Security**

Supervisory Control and Data Acquisition (SCADA) systems are widely used in critical infrastructure such as water treatment plants, power grids, and industrial automation. These systems enable real-time monitoring and control of physical processes through Industrial Control Systems (ICS). However, many SCADA systems rely on legacy communication protocols like Modbus/TCP, which lack built-in security features such as authentication and encryption. This makes them highly vulnerable to cyber threats including unauthorized access, command injection, and network-based attacks.

### **1.2 Problem Statement**

Industrial Control Systems using Modbus/TCP are inherently insecure and susceptible to attacks such as malicious write operations, network scanning, and unauthorized command execution. The absence of proper monitoring and intrusion detection mechanisms makes it difficult to identify and respond to these threats in real time.

### **1.3 Objective of the Project**

- To design and simulate a SCADA-based industrial environment
- To monitor Modbus/TCP communication between PLC and SCADA
- To detect cyber attacks such as scanning and unauthorized writes
- To implement an intrusion detection and monitoring framework
- To generate logs and alerts for analysis and validation

### **1.4 Overview of CS-Shield**

CS-Shield is a defensive monitoring framework designed to secure SCADA networks by providing real-time traffic monitoring, intrusion detection, and centralized log analysis. It integrates tools such as Suricata, Zeek, and Wazuh to detect and analyze suspicious activities in industrial communication.

### **1.5 Scope and Limitations**

The project focuses on a simulated SCADA environment using virtual machines and open-source tools. While it effectively demonstrates detection mechanisms, it does not cover large-scale industrial deployment or advanced AI-based threat detection.

## **CHAPTER 2**

### **CRITICAL REVIEW**

#### **2.1 Existing SCADA Security Approaches**

Traditional SCADA security mechanisms primarily focus on preventive measures such as firewalls, access control systems, and network segmentation. These approaches aim to restrict unauthorized access and isolate industrial systems from external networks. While they provide a basic level of protection, they do not actively monitor internal network traffic or detect ongoing attacks.

#### **2.2 Limitations of Traditional Monitoring**

Conventional monitoring systems lack real-time visibility and are often dependent on manual inspection or static configurations. They fail to detect advanced threats such as stealthy scanning, unauthorized command execution, or protocol misuse. Additionally, they do not provide detailed insights into traffic behavior, making it difficult to identify anomalies.

#### **2.3 Need for Lightweight Monitoring**

Industrial environments require monitoring solutions that are efficient and do not interfere with real-time operations. Lightweight monitoring systems can analyze traffic continuously without affecting performance. Such systems are essential for detecting suspicious activities while maintaining operational stability.

#### **2.4 Improvement through CS-Shield**

CS-Shield improves upon existing approaches by integrating real-time traffic monitoring, intrusion detection, and centralized logging. It combines signature-based and behavioral analysis techniques to enhance detection accuracy. This results in better visibility, faster response, and improved overall security of SCADA networks.

## **CHAPTER 3**

### **DATA HANDLING AND TRAFFIC MONITORING**

#### **3.1 Types of Data**

The system handles multiple types of data, including network traffic, system logs, and intrusion detection alerts. Network traffic mainly consists of Modbus/TCP packets exchanged between PLC and SCADA systems, while logs provide detailed records of system activities.

#### **3.2 Traffic Monitoring using Suricata**

Suricata is used to monitor network traffic in real time. It captures and analyzes packets based on predefined rules, enabling detection of known attack patterns such as scanning and unauthorized write operations.

#### **3.3 Log Handling**

Logs generated by monitoring tools are collected and stored for further analysis. Proper log handling helps in identifying attack patterns, troubleshooting issues, and validating system performance.

#### **3.4 Basic Threat Intelligence (ThreatFox)**

Threat intelligence sources provide information about known malicious activities and indicators of compromise. Integrating such data enhances detection capabilities by enabling the system to recognize known threats more effectively.

## **CHAPTER 4**

### **DETECTION MECHANISM**

#### **4.1 Intrusion Detection using Suricata**

Suricata acts as the core intrusion detection system, continuously analyzing network traffic for suspicious patterns. It compares incoming data with predefined rules to identify potential threats.

#### **4.2 Signature-based Detection**

This approach relies on detecting known attack signatures. It is effective in identifying previously documented threats and provides accurate results when rules are properly configured.

#### **4.3 Rule Configuration**

Custom rules are defined to detect specific attacks such as unauthorized Modbus register writes and malformed packets. These rules determine how the system responds to suspicious traffic.

#### **4.4 Log Monitoring using Wazuh**

Wazuh collects and analyzes logs from various sources, providing a centralized monitoring platform. It helps in correlating events and identifying complex attack patterns.

#### **4.5 Alert Generation**

Alerts are generated when suspicious activity is detected. These alerts enable quick identification and response to potential threats.

## **CHAPTER 5**

### **SYSTEM SETUP**

#### **5.1 Ubuntu VM Environment**

The project is implemented in a virtual environment using Ubuntu Linux, which provides a controlled and isolated setup for testing and experimentation.

#### **5.2 Tools Used**

The system utilizes tools such as Suricata for intrusion detection, Wazuh for log monitoring, Nmap for attack simulation, iperf3 for performance testing, OpenPLC for PLC simulation, and Node-RED for visualization.

#### **5.3 Basic Network Setup**

All components are connected within the same network, allowing communication between PLC, SCADA interface, monitoring system, and attacker machine. Proper configuration ensures smooth operation and effective monitoring.

## **CHAPTER 6**

### **IMPLEMENTATION**

#### **6.1 Installation & Configuration**

All required tools are installed and configured within the Ubuntu environment. Network interfaces and monitoring settings are properly defined.

#### **6.2 Monitoring Setup**

Suricata is configured to monitor network traffic, while Wazuh is integrated to collect and analyze logs generated by the system.

#### **6.3 Attack Simulation (Nmap)**

Nmap is used to simulate network scanning attacks, allowing the system to evaluate its detection capabilities.

#### **6.4 Performance Testing (iperf3)**

iperf3 is used to measure network performance and ensure that monitoring does not significantly impact system efficiency.

## **CHAPTER 7**

### **RESULTS**

#### **7.1 Alerts Generated**

The system successfully generates alerts for various attack scenarios, including network scanning and suspicious Modbus activity.

#### **7.2 Log Observation in Wazuh**

Logs are collected and visualized in the Wazuh dashboard, providing detailed insights into system behavior.

#### **7.3 Detection of Suspicious Activity**

Unauthorized access attempts, malformed packets, and scanning activities are effectively detected by the system.

#### **7.4 Performance Observations**

The system maintains stable performance during monitoring, demonstrating its efficiency and reliability.

## **CHAPTER 8**

### **MONITORING INTERFACE**

#### **8.1 Wazuh Dashboard**

The Wazuh dashboard provides a centralized interface for viewing logs and alerts in real time.

#### **8.2 Log Visualization**

Logs are presented in an organized manner, making it easier to analyze system activity and detect anomalies.

#### **8.3 Event Monitoring**

Real-time event monitoring enables quick identification and response to potential threats, improving overall system security.

## **CHAPTER 9**

### **LIMITATIONS AND FUTURE SCOPE**

#### **9.1 Limitations**

The project is limited to a simulated environment and may not fully represent real-world industrial systems. Signature-based detection may fail to identify unknown threats, and scalability is limited.

#### **9.2 Future Scope**

Future improvements include integrating machine learning techniques for anomaly detection, deploying the system in real industrial environments, and enhancing threat intelligence capabilities.

## **CHAPTER 10**

### **CONCLUSION**

The CS-Shield project demonstrates an effective approach to securing SCADA networks through real-time monitoring and intrusion detection. By integrating tools for traffic analysis, alert generation, and centralized log management, the system enhances visibility into industrial network activities and enables early detection of potential threats. It highlights the importance of implementing layered cybersecurity mechanisms in industrial environments, where even minor vulnerabilities can lead to significant operational risks. Furthermore, the project provides a practical foundation for future research and development, allowing improvements such as advanced anomaly detection, scalability, and integration with real-world industrial systems.

## **CHAPTER 11**

### **DEVELOPMENT PROCESS**

#### **11.1 Implementation Steps**

The development process includes setting up the SCADA simulation, configuring monitoring tools, integrating detection mechanisms, and testing the system.

#### **11.2 Tools Configuration**

Each tool is configured to perform specific tasks such as traffic monitoring, log analysis, and visualization.

#### **11.3 Challenges and Solutions**

Challenges such as configuration complexity and false positives were addressed through careful tuning and testing, ensuring reliable system performance.

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